Math 421

Friday, October 17

Announcements

- No homework due today

$(1, 0, 2, 0, 3, 0, \dots)$

$(1, 2, 3, \dots)$

Theorem 11.2

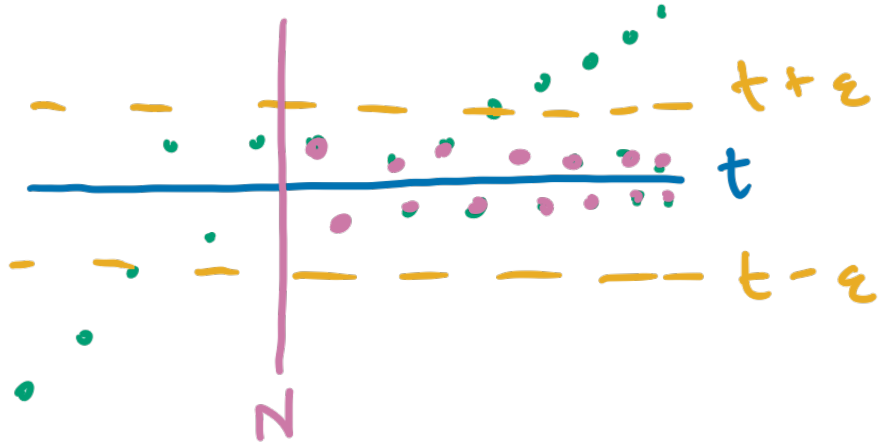
Thm. Let (s_n) be a sequence.

1. If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
2. If the sequence (s_n) is unbounded above, it has a subsequence with limit $+\infty$.
3. If the sequence (s_n) is unbounded below, it has a subsequence with limit $-\infty$

In all cases, the subsequence can be taken to be monotonic.

$(1,0,2,0,3,0,\ldots)$

$(0,0,0,0,\ldots)$



Discussion Questions

- What is part 1 of the theorem saying you your own words?
  - If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is $\boxed{\text{infinite for all } \epsilon > 0.}$

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    set of n such that the distance between sequence elements and t is < ε.

- Convince yourselves the forward direction of part 1 is true: If there is a subsequence of (s_n) converging to t then set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$. If $(t_n) \to t$ then for all $\varepsilon > 0$, there exists N such that for all $\boxed{n > N}$, $|t_n - t| < \varepsilon$.

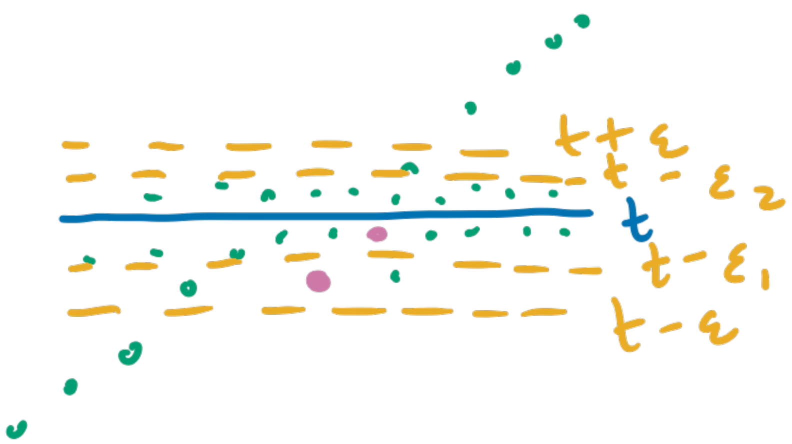
  infinitely many

- Convince yourselves parts 2 and 3 are true.

$s_n = \begin{cases} s_{2k} = t \\ s_{2k+1} = 2k+1 \end{cases}$ $(s_n) = (t)$

$(1, 0, 2, 0, 3, 0, \dots)$

$(0, 0, 0, \dots)$



Theorem 11.2 – Proof of 1

Thm. Let (s_n) be a sequence. If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.

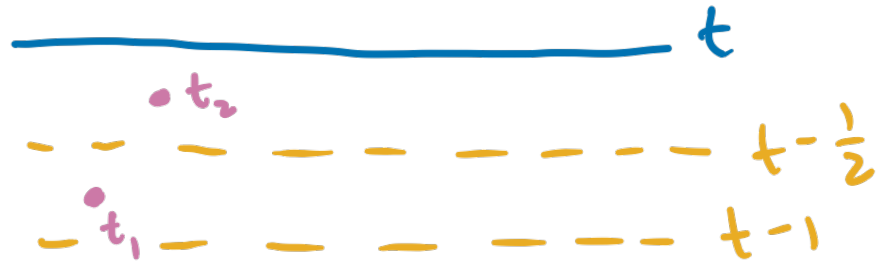
Discussion. $(\Leftarrow)$ Constructive proof.

Case 1: If $s_n = t$ for infinitely many values of n. Construct a subsequence consisting of these terms.

Case 2: If $s_n = t$ for finitely many values of n. So $s_n \neq t$ for infinitely many values of n. Then $\{n \in \mathbb{N} : 0 < |s_n - t| < \varepsilon\} =$ $\boxed{\{n \in \mathbb{N} : t - \varepsilon < s_n < t\}} \cup \{n \in \mathbb{N} : t < s_n < t + \varepsilon\}$ is infinite. (from hypothesis)

So either one must be infinite.

Theorem 11.2 – Proof of 1



Discussion. (cont.)

$\{n \in \mathbb{N} : t - \varepsilon < s_n < t\}$ is infinite for all $\varepsilon > 0$.

- let $\varepsilon = 1$ $\{n \in \mathbb{N} : t - 1 < s_n < t\}$ is infinite. Pick one, n_1,

  set $t_1 = s_{n_1}$

- let $\varepsilon = \frac{1}{2}$ $\{n \in \mathbb{N} : t - \frac{1}{2} < s_n < t\}$ is infinite. Pick one, n_2,

  set $t_2 = s_{n_2}$.

- let $\varepsilon = \frac{1}{k}$ $\{n \in \mathbb{N} : t - \frac{1}{k} < s_n < t\}$ is infinite, Pick one, n_k,

  set $t_k = s_{n_k}$.

since $t - \frac{1}{k} \leq t_k \leq t$ for all k and $\lim(t - \frac{1}{n}) = t$

$\lim t = t$ by the squeeze lemma, $\lim t_n = t$.